RIPPLES IN SPACE

Wavelike behavior is fundamental not just to electromagnetic radiation (see p.14) but also to the quantum behavior of particles. Unlike particles, waves can pass through each other to boost the overall disturbance in some places and decrease it in others (an effect called interference) and also spread into the "shadows" cast by barriers (diffraction). When they encounter a boundary between two different materials, waves can be bounced back (reflection) or slowed down and deflected onto new paths (refraction).

ENERGY TRANSFERENCE

Waves are repeated oscillations (fluctuations) around a fixed midpoint. While waves transfer energy, they do not carry matter from one place to another.